

# Lecture 8 - Cost Curves

ECON 3070 - Intermediate Microeconomic Theory

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# Overview

In the previous lecture, we began talking about costs.

- We considered how output varies as we change the level of inputs (production function).
- And how firms choose the mix of inputs that produces a given quantity at the lowest price (optimal production).
- We considered what happens to the firm's lowest-cost input combination when prices or quantity changed (comparative statics).
- And we looked at how a firm chooses their input combination when the quantity of some input is fixed (short-run production).

# Overview

In this section, we will learn how to calculate total, marginal, and average costs, both in the short run and the long run.

- We will look at these costs graphically, and analyze their relationship.
- We will also introduce concepts such as economies of scale, scope, and experience.

# Long Run Total Cost Curve

In the previous chapter, we learned how to calculate total costs as a function of  $Q$ ,  $w$ , and  $r$ .

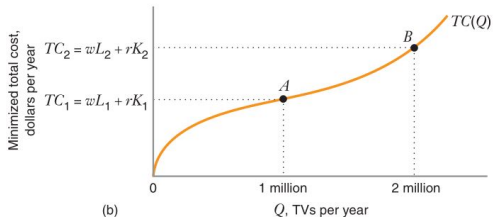
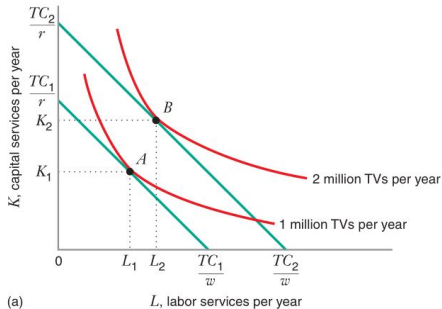
Remember that

$$TC = w * L^*(w, r, Q) + r * K^*(w, r, Q)$$

where  $L$  and  $K$  are the cost-minimizing levels of labor and capital.

- Optimal  $L$  and  $K$  are functions of the wage, the rental rate, and the output quantity.

# Long Run Total Cost Curve



# Long Run Total Cost Curve

Note that the total cost curve is always upward sloping.

- Suppose you could produce 100 units for \$1,000, and 150 units for \$900.
- Then you could also produce 100 units for \$900 ... by producing 150, and throwing some away!

Also note that in the long run,  $Q(0) = 0$ .

- Since all inputs can vary, the cost-minimizing level of all inputs needed to produce 0 units of output is \$0.

# Long Run Total Cost Curve

## Try It Yourself

Suppose a firm has the total cost function  $TC(Q) = \frac{Q}{25}\sqrt{wr}$ . Is the total cost curve increasing with  $Q$ ? Do costs increase for the firm when the cost of capital  $r$  increases?

## Solution

**Is  $TC$  increasing in  $Q$ ?** Take the derivative:

$$\frac{\partial TC}{\partial Q} = \frac{\sqrt{wr}}{25} > 0$$

Yes – since  $w, r > 0$ , this is always positive, so total cost is increasing in  $Q$ .

**Does  $TC$  increase when  $r$  increases?** Take the partial derivative with respect to  $r$ :

$$\frac{\partial TC}{\partial r} = \frac{Q}{25} \cdot \frac{w}{2\sqrt{wr}} = \frac{Qw}{50\sqrt{wr}} > 0$$

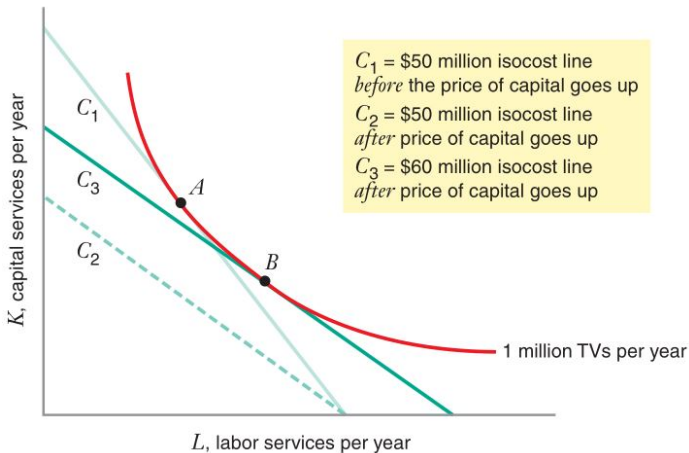
Yes – for any  $Q > 0$ , this is also positive, so total cost rises when the rental rate of capital increases.

# Comparative Statics of Long-run Total Cost

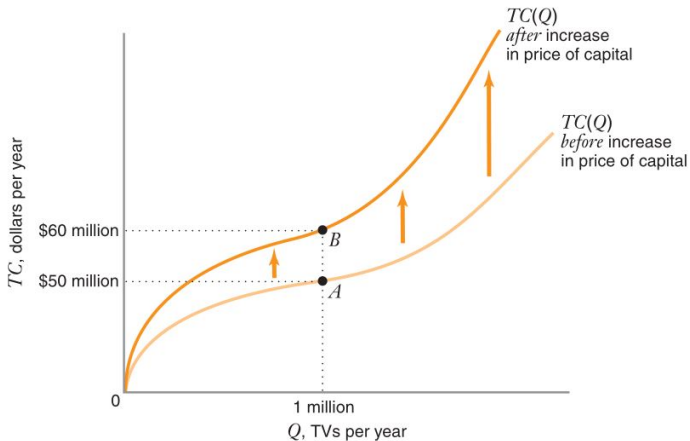
When the price of one input increases, the firm may substitute away from that input, but their TC still increases.

- If TC stayed the same at the higher input price, why wouldn't they have chosen that input combination to start with?
- And in most cases, the increase in cost is greater when a larger quantity is being produced.

# Comparative Statics of LRTC



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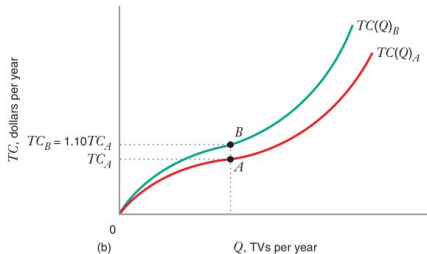
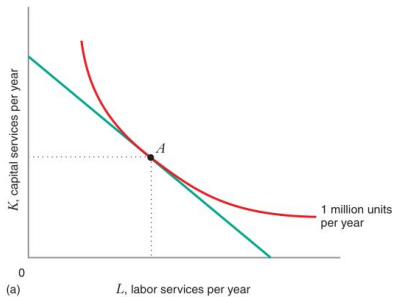
What if both input prices increase proportionally?

Then the slope of the isoquants ( $-w/r$ ) doesn't change.

- So the point of tangency doesn't change.
- In other words, the optimality condition still holds.

⇒ The input combination doesn't change. It just gets proportionally more expensive to make.

# Comparative Statics of LRTC



# Long-Run Average and Marginal Costs

There are two other important costs we need to consider: *marginal cost* and *average cost*.

- Marginal cost is important in deciding *how many* units to produce.

**How much should the firm produce?**

- Average cost is useful in determining a firm's overall profitability (should they produce anything at all).

**Should they be producing at all?**

# Long-Run Average and Marginal Costs

**Long-run average cost** is the firm's cost per unit of output.

$$AC(Q) = \frac{TC(Q)}{Q}$$

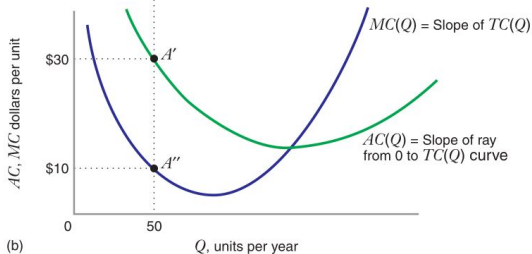
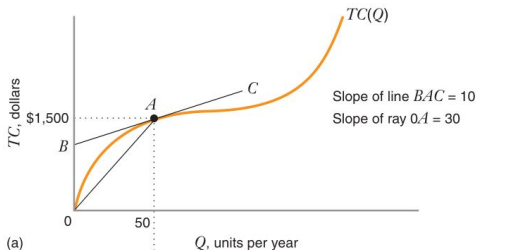
Also the slope of a ray from the origin to the point on the total cost curve.

**Long-run marginal cost** is the rate at which long-run total cost changes with respect to a change in output.

$$MC(Q) = \frac{\Delta TC}{\Delta Q}$$

Also the slope of the tangent line at that quantity.

# Long-Run Average and Marginal Costs



# Long-Run Average and Marginal Costs

**Try It Yourself** Suppose that the total cost function for a cost-minimizing firm is given by  $TC(Q) = 4Q$ . What is the firm's marginal and total cost?

**Solution** The total cost function is simply  $TC(Q) = 4Q$  – so total cost at any quantity  $Q$  is  $4Q$ .

Marginal cost is the derivative of  $TC$  with respect to  $Q$ :

$$MC(Q) = \frac{dTC}{dQ} = 4$$

Marginal cost is **constant at \$4**, regardless of quantity produced.

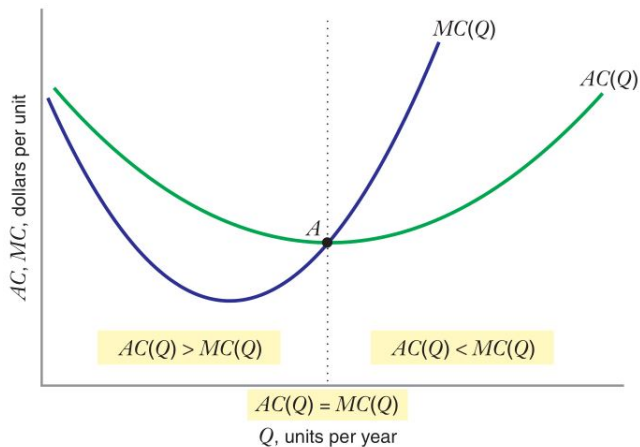
Note that average cost is  $AC(Q) = TC(Q)/Q = 4$  as well – since  $TC$  is linear through the origin (no fixed costs),  $AC = MC = 4$  at every quantity.

# Relationship Between AC and MC Curves

As we saw with marginal product and average product, there is a systematic relationship between marginal cost and average cost.

- If average cost is decreasing as quantity is increasing, then average cost is greater than marginal cost.
- If average cost is increasing as quantity is increasing, then average cost is less than marginal cost.
- If average cost is constant as quantity is increasing, then average cost equals marginal cost.

# Relationship Between AC and MC Curves



# Economies of Scale

It's important to consider what will happen to a firm's production costs as they produce more output.

- Perhaps as a firm produces more output, employees can specialize more...
- ... or maybe their production process uses *indivisible inputs*, which cannot be scaled down easily...
- ...and their cost per unit falls.

A firm is said to experience **economies of scale** when average cost of production falls as output rises.

# Economies of Scale

Or perhaps as a firm grows, managerial needs outpace output growth increase the per-unit cost.

- When costs arise due to an increase in managerial needs, this is known as **managerial diseconomies**.
- When average cost rises as output rises, a firm is said to experience **diseconomies of scale**.

# Economies of Scale

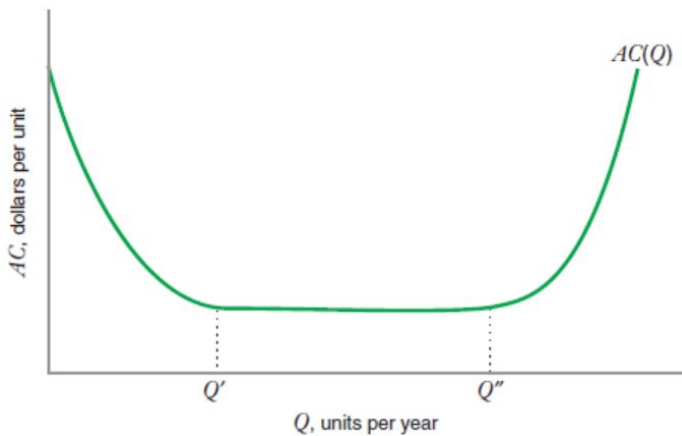
A firm is said to be producing at its **minimum efficient scale (MES)** when long-run average cost is at its minimum.

- This is an important concept since it tells us, *at best*, how efficiently firms can produce the good

Firms with a large minimum efficient scale relative to market size have greater economies of scale.

- In these markets, it is more difficult for smaller firms to compete.

# Economies of Scale



# Economies of Scale and Returns to Scale

If average cost *decreases* as output *increases*, then we have *economies of scale* and *increasing returns to scale*.

- E.g.  $Q = L^2$

If average cost *increases* as output *increases*, then we have *diseconomies of scale* and *decreasing returns to scale*.

- E.g.  $Q = \sqrt{L}$

If average cost *stays the same* as output *increases*, then we have *neither economies nor diseconomies of scale* and *constant returns to scale*.

- E.g.  $Q = L$

# Economies of Scale and Returns to Scale

Note that the previous slide assumed that input price didn't depend on quantity of production.

A firm could experience constant returns to scale in technology...

- ...but if quantity discounts are available for inputs, they may experience economies of scale.
- Or if an input is scarce, the price may rise as output increases and that same firm may experience diseconomies of scale.

# Measuring Economies of Scale

Can use elasticities to tell us how sensitive total cost is to output.

The **output elasticity of total cost** is given by:

$$\epsilon_{TC,Q} = \frac{\frac{\Delta TC}{TC}}{\frac{\Delta Q}{Q}} = \frac{\frac{\Delta TC}{\Delta Q}}{\frac{TC}{Q}} = \frac{MC}{AC}$$

| Value of $\epsilon_{TC,Q}$ | MC Versus AC | How AC Varies as Q Increases | Economies/<br>Diseconomies of Scale |
|----------------------------|--------------|------------------------------|-------------------------------------|
| $\epsilon_{TC,Q} < 1$      | $MC < AC$    | Decreases                    | Economies of scale                  |
| $\epsilon_{TC,Q} > 1$      | $MC > AC$    | Increases                    | Diseconomies of scale               |
| $\epsilon_{TC,Q} = 1$      | $MC = AC$    | Constant                     | Neither                             |

# Measuring Economies of Scale

## Try It Yourself

Which of the following is a possible reason why a firm might experience increasing returns to scale, and diseconomies of scale?

- A) The firm uses indivisible inputs in its production process.
- B) The firm experiences managerial diseconomies when output rises.
- C) The firm receives quantity discounts on its inputs when output rises.
- D) The firm's employees are able to specialize more when output rises.

## Solution

- **A)** False. Indivisible inputs create *economies* of scale.
- **B) True.** Managerial diseconomies can raise average cost even if the production technology exhibits increasing returns to scale.
- **C)** False. Quantity discounts reinforce economies of scale.
- **D)** False. Specialization also creates economies of scale.

**Key idea:** Returns to scale describe the *production technology*, while economies of scale depend on *average cost*, which can also be affected by managerial and organizational costs.

**Answer: B**

# Short-Run Cost Curves

In the previous lecture, we learned the difference between the short run and the long run.

- In the short run, the levels some inputs cannot be varied.
- In the long run, the levels of *all* inputs can.
- In this section, we will assume that capital ( $K$ ) is the fixed input.

# Short-Run Cost Curves

The short-run total cost curve  $STC(Q)$  tells us the minimized total cost of producing  $Q$  units of output when at least one input is fixed.

The total cost curve can be divided into a **total variable cost curve (TVC)** and a **total fixed cost curve (TFC)**.

- $TVC(Q)$  is the sum of expenditures on variable inputs at the cost-minimizing input combination.
- $TFC(Q)$  is the total cost of fixed inputs.

$$STC(Q) = TVC(Q) + TFC(Q)$$

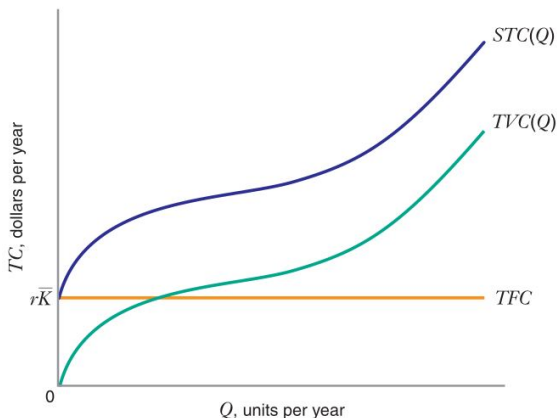
# Short-Run Cost Curves

Because we let capital be our fixed input,  $TFC$  is simply the amount of money spent on  $\bar{K}$  units of capital. In other words,  $TFC = r\bar{K}$ .

Thus,  $STC(Q) = TVC(Q) + r\bar{K}$ .

# Short-Run Cost Curves

Note that the vertical distance between the  $STC(Q)$  curve and the  $TVC(Q)$  curve is equal to  $r\bar{K}$ .



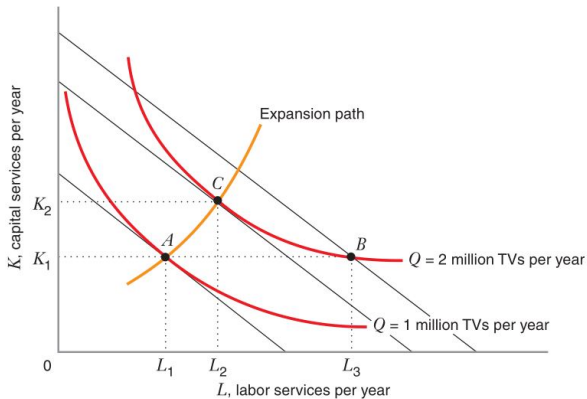
# Relationship Between $LTC$ and $STC$

Consider a firm that produces televisions, and wants to expand production.

- In the short run, they can only adjust their level of labor.
- But in the long run, they can adjust the level of capital as well.

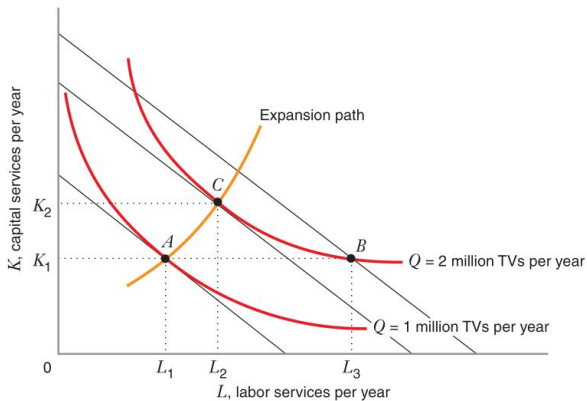
# Relationship Between $LTC$ and $STC$

Going from 1 million to 2 million TVs. In the short run, you have to go to  $B$ .



# Relationship Between $LTC$ and $STC$

Going from 1 million to 2 million TVs. In the short run, you have to go to  $B$ . In the long run, you can go to  $C$  at a lower cost.



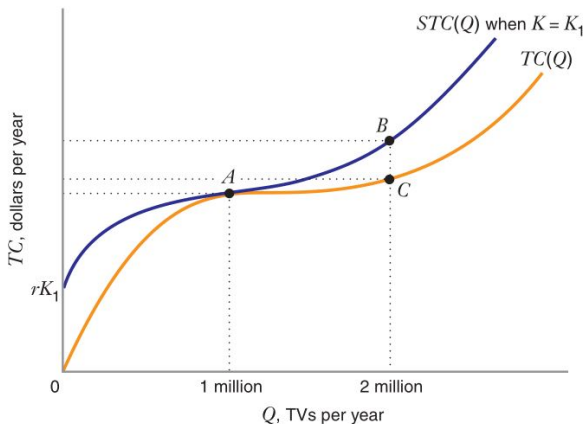
## Relationship Between $LTC$ and $STC$

Note that total cost will **never** be lower in the short run than in the long run.

- If you can use that input combination in the short-run, you can use it in the long run!

# Relationship Between $LTC$ and $STC$

Below are the  $STC$  and  $LTC$  curves for some firm. the  $STC(Q)$  always lies above the  $LTC(Q)$ .



# Short-Run Average and Marginal Cost Curves

We can define **short-run average cost**,  $SAC$ , and **short-run marginal cost**,  $SMC$ , similarly to their long-run counterparts.

$$SAC(Q) = \frac{STC(Q)}{Q} \text{ and } SMC(Q) = \frac{\Delta STC}{\Delta Q}$$

We can also break  $SAC(Q)$  into average variable costs and average fixed costs.

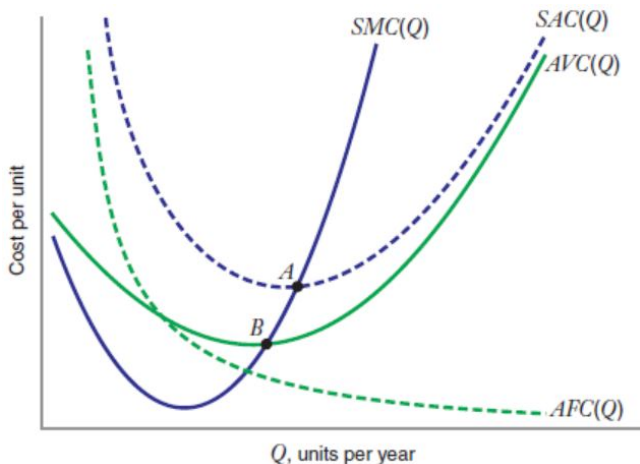
$$SAC(Q) = AVC(Q) + AFC(Q)$$

# Short-Run Average and Marginal Cost Curves

If we plot these curves, we can see that  $SAC$  is found by vertically summing the  $AFC$  and  $ATC$  curves.

- The  $AFC$  curve slopes downward because fixed costs do not change as output increases. The fixed cost is spread over more and more units of output.
- Note that the marginal cost curve intersects the average variable and average total cost curves at their minima.

# Short-Run Average and Marginal Cost Curves



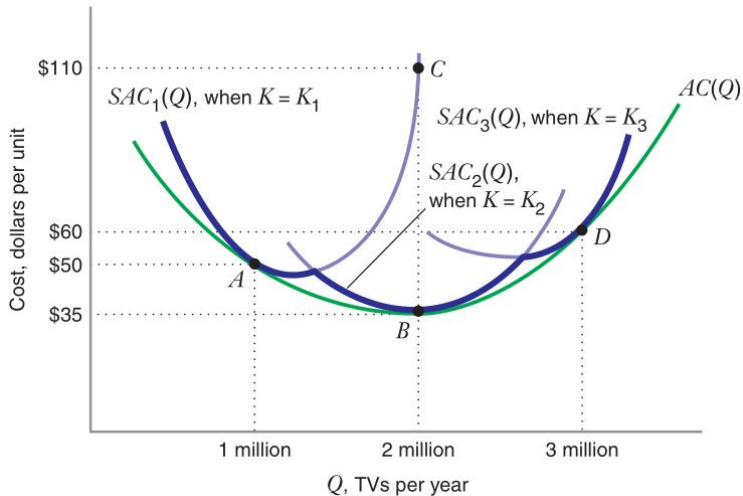
# Relationship Between Long-Run and Short-Run $AC$ and $MC$ Curves

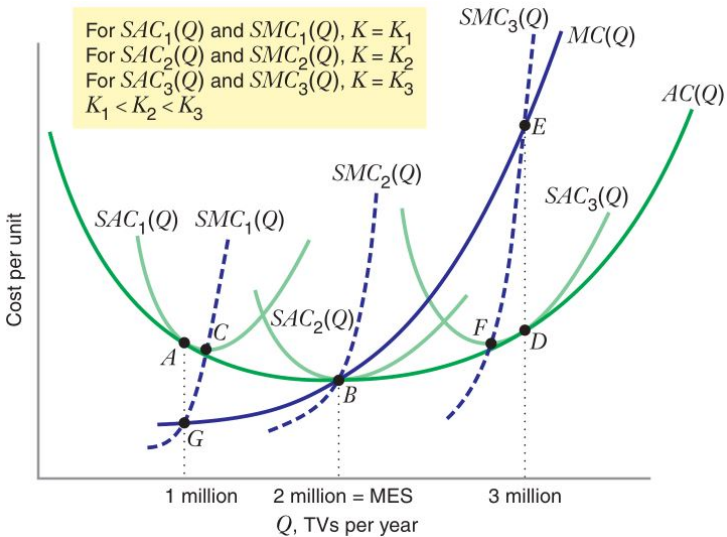
A firm can have many short-run average cost curves depending on the level of capital.

In the long-run, capital can vary

- And the firm will choose the level of capital that minimizes average cost for a given output level.
- Thus, the long-run average cost curve is the boundary of all of the short-run average cost curves.
- It traces the minimum SAC for every level of output.

The long-run curve can be thought of as the lower envelope of the set of short-run curves.





# Relationship Between Long-Run and Short-Run $AC$ and $MC$ Curves

## Try It Yourself

Which of the following relationships is always true?

- A)  $MC(Q) > SMC(Q)$
- B)  $AC(Q) > MC(Q)$
- C)  $SAC(Q) > MC(Q)$
- D)  $SAC(Q) > AC(Q)$

## Solution

- **A)** False.  $MC$  and  $SMC$  cross, so either can be larger.
- **B)** False.  $MC < AC$  when  $AC$  is falling, but  $MC > AC$  when  $AC$  is rising.
- **C)** False.  $MC$  and  $SAC$  also cross.
- **D) True.** The long-run average cost curve is the *lower envelope* of all short-run average cost curves, so

$$SAC(Q) \geq AC(Q),$$

with equality only at the output where the fixed capital level is optimal.

**Answer: D**

# Other Determinants of Cost

So far, we have focused on firms that only produce 1 good or service.

- But many firms sell a large variety of goods (Samsung, Johnson & Johnson, General Electric).

Why do some firms choose to produce many different products?

- Often, because fixed costs can be spread among many more units.
- For example, a satellite TV company broadcasts many channels...  
Not just one.

# Other Determinants of Cost

Or Budweiser can use their machinery to bottle many different types of beer.

- Firms can also spread administrative costs over more units of output
  - E.g. the accounting department or human resources department.

Can also receive discounts on inputs (if their products use a common input) or shipping costs.

## Other Determinants of Cost

Mathematically, a firm is said to experience **economies of scope** if:

$$TC(Q_1, Q_2) < TC(Q_1, 0) + TC(0, Q_2)$$

In other words, the total cost of 1 firm producing both goods ( $Q_1$  and  $Q_2$ ) is less than the sum of the total costs of each firm producing separately

The opposite could be true, implying **diseconomies of scope**

## Other Determinants of Cost

Sometimes costs can fall over time as a firm learns more efficient production methods.

This is known as **economies of experience** or **learning-by-doing**.

Think of the solar industry.

- The price of solar power has fallen 60% in the last decade.
- This is at least partly a result of accumulated experience.